Sylvester: A Benchmark and Empirical Study of Lifecycle-Aware Entity Resolution on Open Financial Data

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ABSTRACT

We introduce Sylvester, a public benchmark for *lifecycle-aware* entity resolution on financial entities: matching records across mergers, rebrands, spinoffs, and jurisdiction migrations where the surface form has deliberately changed. Built from public GLEIF Level-1/2, SEC EDGAR, and Wikidata, it provides 125,519 labeled pairs over 3.3M entities across three held-out evaluation regimes (time, event type, jurisdiction), with a three-pool evaluation protocol whose job is to separate methods that learn the pool from methods that learn the task: the released pool, a pool from an unrelated miner, and candidates retrieved from the full universe. The protocol has teeth: the benchmark deliberately mines name-similar negatives against name-different positives, and the methods that exploit this — inverted name similarity (64–70% Hits@1 on time/event) and a name-feature GBM (0.73–0.84, the best name-input scores we observe) — collapse on the other two pools. Among methods restricted to name and registration-state inputs, a 22M-parameter MiniLM cross-encoder is the only one that transfers: 0.82–0.87 AUROC across the three regimes, 57–61% Hits@1 on time/event (±2.0–2.3 pp 95% CI), 24–35 pp above the strongest bi-encoder, and, under a uniform single-pair protocol, above all six zero-shot LLMs tested (four organizations) on the two large splits. But the binding constraint is the input more than the architecture: given the *full* structured registry record (liveness, addresses, previous names, dates — fields no name-only method sees), a gradient-boosted matcher transfers across every pool, beating the name-input cross-encoder on-pool by 18 pp ($p < 10^{-4}$) at negligible cost; architecture then decides what survives shift — a cross-encoder over the same record ties the trees on curated pools and wins where deployment lives, +12 pp retrieval-conditional on the fielded universe ($p < 10^{-4}$), the best we measure. The lifecycle signal is in the record; joint encoding keeps most of it under pool shift. End-to-end even the best system reaches 21–39% Hits@1 because Recall@20 caps at 33–50%: the bottleneck is retrieval.

KEYWORDS

entity resolution, record linkage, financial entities, lifecycle events, GLEIF, benchmark, cross-encoder

1 INTRODUCTION

Twitter Inc. became X Corp. Facebook became Meta. Sprint was absorbed into T-Mobile. Pipelines silently breaking on such transitions is a familiar failure mode across fintech engineering, quant research, and compliance. The records look completely different on the surface — different legal names, different tickers, sometimes different jurisdictions and legal forms — yet they describe the same continuing economic entity.

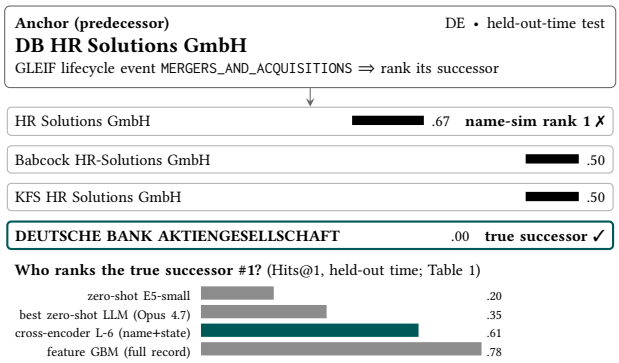


Figure 1: The task and the headline result. *Top:* a real held-out-time test anchor (4 of 7 candidates; bars = token Jaccard). Methods must rank the true successor above deliberately name-similar negatives; name similarity ranks a coincidental collision first and the true successor — zero shared tokens — last (§6.1). *Bottom:* on this pool the gap closes with *input* more than architecture: zero-shot name scoring fails, a cross-encoder on name+state reaches .61, boosted trees over the full record reach .78 (a name-only exploit hits .77 here but collapses off-pool; Figure 2, §5.4).

Pair-matching ER — including LLM-based methods that achieve 99% F1 on point-in-time records [10] — treats each record as a snapshot of identity. Temporal entity-linking benchmarks such as TempEL [13] address mention-to-KB disambiguation against an evolving knowledge base, a different task. Financial KG construction [1] works at a single time slice. To the best of our knowledge no public benchmark targets the specific question we care about: *given two records observed at different times, do they describe the same economic entity, accounting for any lifecycle event that may have occurred between them?*

A natural concern: the benchmark is engineered against name similarity — negatives are mined name-similar while successors are, by definition, name-different. This is deliberate, documented, and measured: the inverted exploit is its own baseline (§5.4). The benchmark asks the discriminative sub-problem after a name-aware retriever narrows the field: *given a name-similar candidate pool, what identifies the true lifecycle successor?* (Figure 1.)

Contributions.

(1) **An open benchmark** (§3): 125,519 labeled pairs over 3.3M GLEIF Level-1 entities, hard negatives in four classes (the parent/subsidiary class drawn from GLEIF Level-2 relationship records), and three held-out regimes (time, event type,

jurisdiction), with construction choices and their scope documented throughout.

(2) **A three-pool diagnostic, and what it separates** (§5): negatives are mined name-similar while positives are name-different by construction — not a claim about ER generally — so *inverted* name similarity and a name-feature GBM post the best name-input on-pool scores we observe, then collapse on the two other pools (§5.4). Among name+state-input methods a 22M-parameter cross-encoder is the only one that transfers (24–35 pp Hits@1 over the best bi-encoder, above all six zero-shot LLMs on TIME/EVENT; gains concentrated on M&A and absorption). But given the *full* structured record, a gradient-boosted matcher transfers everywhere too and beats the cross-encoder on-pool by 18 pp ($p < 10^{-4}$), while a cross-encoder over the same record wins the fielded pool by 12 pp (§5.4). The transferable signal is in the record's structured fields — no name-only method can recover it — and joint encoding keeps the most of it under shift.

(3) **The deployment bottleneck is retrieval, not reranking** (§5.6): fielded over 3.3M entities, end-to-end Hits@1 falls to 15–39% because Recall@20 caps at 33–50%, consistent with the blocking literature's focus on candidate generation [12]. The rerankers themselves are effective (46–78% conditional Hits@1); the system breaks at the first stage.

Baseline comparisons use the single fine-tuned L-6; the enriched L-6×L-12 ensemble adds 9–11 pp (Table 2).

2 RELATED WORK

Classical ER [4] matches records via similarity over structured fields; modern PLM cross-encoders on serialized records (Ditto [8], GraLMatch [3]) are now standard, and the OpenSanctions Pairs corpus [10] benchmarks LLM matching at 755K pairs — all under a *snapshot* framing, not lifecycle-spanning matches. Temporal record linkage [7] pioneered matching drifting entities, predating modern encoders and open registries. AnyMatch [14] shows small fine-tuned models rival LLM matchers at a fraction of the cost, and WDC Products [9] studies how hard-negative construction shapes benchmark difficulty — the design axis §5.4 interrogates here. TempEL [13] time-stratifies entity linking against an evolving Wikipedia, but that task is mention-to-KB disambiguation, not record-to-record matching over legal entities. Financial-filing knowledge graphs (FinReflectKG [1], from this conference) are constructed at a single time slice; the lifecycle-bridge question is orthogonal.

3 THE SYLVESTER BENCHMARK

3.1 Data sources

We construct Sylvester entirely from open, persistently-licensed sources:

- **GLEIF Level-1 LEI-CDF.** The LEI repository (3.3M entities, 2026-05-19 snapshot), CC0, refreshed every eight hours. Each `LEIRecord` carries legal name, jurisdiction, legal form, addresses, registration metadata, a dated `LegalEntityEvents` log, and a `SuccessorEntity` field linking predecessor to successor by LEI.
- **GLEIF Level-2 Relationship Records.** 473,839 typed inter-LEI relationships, used in hard-negative construction.
- **Wikidata.** High-profile M&A via SPARQL on P156 (*followed by*); a marquee supplement and cross-validation source.
- **SEC EDGAR.** `company_tickers.json` (10,354 records) gives a name↔CIK crosswalk to GLEIF for the U.S. public subset (3,334 confident links); released with the benchmark, not used in the evaluation.

3.2 Positive labels

Our gold-standard positives are GLEIF `SuccessorEntity` links: explicit predecessor-LEI → successor-LEI pairs maintained by GLEIF's network of Local Operating Units. The 2026-05-19 snapshot contains 50,202 records with a non-null `SuccessorEntity` element; 33,645 of these have a resolvable `SuccessorLEI` (the rest reference successors that never registered an LEI and are excluded).

Label quality validation. A five-class auto-classifier on the successor pairs (using documented event types, normalized name delta, jurisdiction, legal form) yields: `strong_real` (documented event + name change) 25,428; `likely_real_name_changed` 3,330; `likely_real_event_only` 1,704; `ambiguous` 1,355; **`likely_administrative`** 1,828 (5.4%, filtered out). We retain the first four classes (94.6%; `ambiguous` stays because every link is still LOU-curated), yielding 31,817 plausibly-real predecessors (31,824 positive pairs; some predecessors have multiple successors).

3.3 Hard negatives

For each positive predecessor we mine up to 80 hard negatives drawn from four classes:

- `parent_subsidiary` (221 pairs) — LEIs that share a Level-2 relationship-record edge (consolidation, fund-management, sub-fund) with the predecessor. These look related but are legally distinct.
- `same_name_same_country` (10,208) — distinct LEIs with identical normalized legal names in the same jurisdiction: coincidental name collision vs. true lifecycle linkage.
- `same_name_diff_country` (10,703) — distinct LEIs with identical normalized name in a different country.
- `token_overlap_same_country` (232,601) — fuzzy fallback: distinct LEIs in the same country sharing ≥ 2 content tokens after normalization, ranked by Jaccard. Covers the long tail when name-exact collisions are sparse.

After deduplication: 253,733 candidate (predecessor, hard-negative) pairs covering 20,192 of 31,817 predecessors (≈63%). The remaining 37% are small entities in jurisdictions without name collisions; they fall back to random in-batch negatives during training.

3.4 Evaluation splits

For each positive we identify its *anchor lifecycle event* — the highest-priority COMPLETED event on the predecessor's record

by type-priority then effective date. We then assign each positive (and its associated negatives) to one of three split regimes:

- **Held-out time:** anchor effective date < 2025-01-01 (train) vs. ≥ 2025-01-01 (test); completed-event volume rises ≈4× in 2022 when GLEIF tightened reporting, so the effective window is 2022–2026. Test: 3,448 positives, 2,007 Hits@1-eligible anchors (§5.2).
- **Held-out event type:** train on M&A, rename, legal-form change; test on spinoff, demerger, breakup, absorption — dominated by *absorption* (2,609 of 2,760 anchors, 94.5%), the rest statistically thin (§5.5).
- **Held-out jurisdiction:** train on US + GB + DE; test on SG + JP + BR. Test composition: 161 anchors, 161 positives, of which 88 are Hits@1-eligible — the smallest split; its headline numbers carry ∼±8.5 pp 95% bootstrap CIs (vs. ∼±2.0–2.3 pp on time/event).

We refer to these regimes as TIME, EVENT, and JURIS. Each positive gets up to 5 hard negatives if its splits are all `train`, or up to 10 if any split places it in `test`, yielding 125,519 final pair rows.

Wikidata marquee supplement. Released alongside the benchmark: 302 high-profile M&A pairs from Wikidata P156 both-sides-resolved to GLEIF, 283 not captured by GLEIF `SuccessorEntity` (Sprint → T-Mobile, Wyeth → Pfizer, Interbrew → AB-InBev); not used in the headline evaluation.

4 METHOD

We present two methods at different cost-accuracy points: a bi-encoder (§4.1) that embeds each side independently and scores by cosine — cheap, but hemmed in by the benchmark's anti-correlation property (name similarity anti-predicts the label here; §5.4) — and a cross-encoder (§4.2) that reads both records together, 4–10× more expensive per pair, producing the headline name-input result.

4.1 Bi-encoder

We fine-tune `intfloat/e5-small-v2` (33M params, 384-dim) via supervised contrastive learning with state features folded into the text prompt: "`query:` n_ℓ `| jurisdiction:` c_ℓ `| form:` f_ℓ". Training uses the 26,297 positives in the strict no-leakage intersection of all three split regimes (one hard neg per anchor), MultipleNegativesRanking loss (InfoNCE; in-batch + explicit hard negs), 3 epochs, batch 32, AdamW lr 2×10^{-5}, ≈6 min on Apple silicon. Ablation variants in Table 1: `method_no_hard` (in-batch negs only), `method_v2_full` (adds E5-cosine hard negs + 1,372 transitive positives). Loss is contrastive; the cross-encoder below uses BCE on the logit — different objectives by architectural convention, not strictly controlled.

4.2 Cross-encoder

A bi-encoder must hold each record at a fixed point in embedding space while the event deliberately changes the record's surface form — it must keep cosine-close two records that look nothing alike. A cross-encoder reads both records jointly and can compare specific tokens between them.

We fine-tune a second-stage cross-encoder that ingests $(\text{record}_A, \text{record}_B)$ as a single concatenated sequence and emits a single relevance logit. The backbone is the 22M-parameter `ms-marco-MiniLM-L-6-v2` cross-encoder from HuggingFace; training uses the same positives as `method_v2_full` (26,297 plus 1,372 transitives) plus a 4:1 sample of E5-cosine hard negatives per positive, with positives labeled 1.0 and negatives 0.0. Loss: standard binary cross-entropy on the logit. 2 epochs, batch 32, lr 2×10^{-5}, 100 warmup steps. The extensions in §5.3 hold architecture/optimizer constant and modify only the serialized text, the training positives, and the backbone.

Inference cost. Scoring all 125,519 pairs takes ≈80 seconds on MPS; in deployment the cross-encoder re-ranks a retriever's top-K, and the benchmark's fixed 5–10-candidate pools make direct full scoring feasible (Table 1).

Hard-negative re-mining. Re-mining with E5-cosine top-20 same-country candidates, in place of the noisy token-overlap fallback, yields 652,764 negatives across 31,714 predecessors (vs. 253,733 over ≈20K before). The cleaner negatives are what the cross-encoder trains on, to large effect — and half of what unlocks the full-record GBM's transfer (§5.4); for the bi-encoder they change nothing (§5.7).

5 EXPERIMENTS

5.1 Baselines

We compare against four similarity baselines — `exact_name` (normalized name equality), `token_set` (token Jaccard), `tfidf_char` (char-3gram TF-IDF cosine), and `sentence_encoder` (zero-shot E5-small-v2, the method's backbone un-tuned) — and one *supervised* classical matcher, `features_gbm`: gradient-boosted trees over twelve string-similarity and field-agreement features in the Fellegi–Sunter lineage [4], trained on the released pool's no-leakage rows; the classical name-feature default (§5.4 upgrades it to the full record). Six zero-shot LLMs run under one protocol: one call per pair, byte-identical prompts carrying the same name+state inputs as the neural models (legal name, jurisdiction, legal form), a 0–10 integer rubric, `temperature=0` where exposed. Five API models from three organizations cover the full test set (26,564 pairs each) — **Claude Haiku 4.5**, **Sonnet 4.6**, **Opus 4.7**, **Gemini 3.1 Pro**, **DeepSeek V4 Pro** (thinking disabled) — and **gpt-oss-20b** (a fourth organization) runs locally on a 56/52/50-anchor sample (Table 1 footnotes).

Paired permutation tests on per-anchor expected Hits@1 separate the comparisons cleanly: the L-6 beats all five API models on TIME and EVENT at $p < 10^{-4}$ (+15 to +37 pp). On the 88-anchor JURIS split it leads Haiku, Gemini, and DeepSeek ($p \leq 0.018$), is within noise of Sonnet ($p = 0.14$), and is indistinguishable from Opus 4.7, which holds the higher point estimate (0.767 vs. 0.750, $p = 0.74$); the ensemble beats Opus on TIME/EVENT ($p < 10^{-4}$), JURIS a coin flip. The panel (Table 1) is strikingly vendor-heterogeneous — every Claude model outscores both other API vendors' frontier models on every split, including Haiku, the cheapest tested; §7 examines the elicitation-format confound behind part of this spread. Under Holm correction across all ≈30 paired tests in the paper, every $p < 10^{-4}$ result survives; the

$p \approx 0.02$ JURIS-level and on-pool-TIME margins do not, and should be read as directional.

5.2 Metrics

We report per-anchor Hits@1/MRR and pairwise AUROC; Hits@3, AUPRC, and threshold-tuned F1 ship with the released metrics. Discrete scorers tie frequently, so Hits@k/MRR are exact *expectations over uniformly random tie-breaking* (a fixed tie order is biased). Per-anchor metrics cover *eligible* anchors — pool size ≥ 2 with the gold present; the 37% of predecessors with no mined hard negative have singleton pools and are excluded (1,612 of 2,760 EVENT-test anchors are eligible; §3).

5.3 Headline results

Table 1 reports test-set metrics across the three split regimes for all baselines and method variants on the released benchmark pool (`pairs.parquet`; token-overlap-mined negatives). Headline cross-encoder Hits@1 95% bootstrap confidence intervals are ±2.0 pp (TIME), ±2.3 pp (EVENT), and ±8.5 pp (JURIS, n=88 anchors); the jurisdiction-split gain comfortably clears its CI band but absolute numbers on that split should be treated as directional. We retest the headline cross-encoder on a mechanically different candidate pool (§5.6) and find the gap shrinks but does not collapse; Table 2 extends the L-6 cross-encoder to the L-6×L-12 ensemble.

Extensions: enriched prompt + L-6×L-12 ensemble. Stacking prompt enrichment (GLEIF Level-2 parent/child names, founding year, ISO country, category), 259 Wikidata-mined training positives, multi-seed averaging, and an L-12 ensemble partner lifts Hits@1 +9 to +11 pp on time/event (+3 on jurisdiction, inside its CI; Table 2) to 0.700 / 0.677 / 0.784; baseline comparisons keep the single L-6 apples-to-apples. One negative result: symmetric scoring (averaging $s(a,b)$, $s(b,a)$) hurts −0.9/−1.9/−9.1 pp — consistent with a *directional* predecessor→successor representation (the TIME delta sits within its CI).

Backbone size does not break the bi-encoder ceiling. E5-small (33M) → E5-base (110M) leaves TIME/EVENT unchanged (−0.2/−1.2 pp; JURIS +2.3): on this one scaling step, cosine-close embeddings for lexically-different pairs are not a problem more parameters fix.

5.4 The construction cuts both ways: an anti-similarity baseline

`token_set`, `tfidf_char`, and `sentence_encoder` all land below 0.5 AUROC on the held-out-time and held-out-event-type splits (Table 1). The cause is mechanical: hard negatives are deliberately name-similar to the predecessor while positives are by construction name-different, so on this candidate distribution name similarity is *evidence against* a true successor link. So we report the mirror image as its own baseline: `anti-token_set` ranks candidates by *negated* Jaccard, and on the curated pool it is strong (Table 1; *above* the single L-6 cross-encoder on time and event) because picking the least name-similar candidate exploits the construction.

The supervised name-feature `features_gbm` is the exploit's learned apotheosis: it discovers the inversion plus the miners' residual signatures, posts the best name-only scores in the paper (Table 1; above the ensemble at $p < 10^{-4}$ on TIME/EVENT), and collapses off-pool (Figure 2; JURIS: 0.36 on Sylvester-TF-IDF, 0.12 fielded; AUROC 0.54–0.57 and 0.45–0.52).

Is that collapse a fact about feature models, or about what we fed this one? Its two handicaps — v1 negatives, name-only features — are removable, separately and jointly. On the cross-encoder's own E5-mined negatives, the name-feature GBM improves cross-pool but still dies fielded (0.25 / 0.25 retrieval-conditional vs. 0.46 / 0.50); a TF-IDF-negative retrain behaves alike (Figure 2, row 3). Given the full structured record (addresses, previous names, liveness, registration status, dates, registrar; the successor fields the labels derive from are excluded) but v1 negatives, it wins the released pool (0.87 / 0.84) yet still fails to cross pools (0.38 / 0.35 on TF-IDF). Given *both*, it transfers everywhere and wins (`features_gbm-full`): 0.78 / 0.75 on-pool (+18 pp over the L-6, $p < 10^{-4}$; +8 pp over the ensemble), 0.78 / 0.75 on TF-IDF, and 0.51–0.66 fielded retrieval-conditional, where the enriched ensemble leads by 1–3 pp on TIME/EVENT and trails by 6 pp on JURIS (all n.s.) — at negligible inference cost. Permutation importance credits fields no name-only method sees: candidate liveness, shared postal/city, previous-name overlap, same registrar. Nor is the transfer a live-vs-dead prior: ablating all four liveness/registration-status features costs 4–7 pp and transfer survives (0.72 / 0.72 on TF-IDF; 0.43 / 0.47 fielded-conditional, at parity with the L-6).

Completing the grid: a cross-encoder trained with the identical protocol but a serialized *full-record* prompt (the same fields the GBM sees) lands 2–5 pp behind the trees on-pool (0.76 / 0.71, $p \leq 0.018$), ties them on TF-IDF (0.77 / 0.73), and pulls decisively ahead where deployment lives: 0.64 / 0.66 / 0.78 fielded retrieval-conditional, +12 pp over the GBM and +9–12 pp over the ensemble (all $p < 10^{-4}$); best end-to-end in the paper (21–39%).

The claims accordingly split by input class and by pool. Among name-input methods — every LLM, bi-encoder, string baseline, and name-feature GBM — the L-6 is the only one whose worst pool stays strong. Given the full record, trees and joint encoding tie on curated pools and the cross-encoder wins the fielded universe. The transferable signal lives in the structured record — no name-only method, joint attention included, can recover fields it was never given — and joint encoding is the best way we test of keeping that signal under distribution shift.

The unsupervised exploit shows the same pattern at lower amplitude, collapsing under every pool change we test, though on one event type it outranks both cross-encoders. On Sylvester-TF-IDF it collapses to chance (0.25 / 0.25 / 0.20 Hits@1) while the cross-encoders hold 0.53–0.66 (Table 4); on the fielded pool it drops to 9–18% (gold-included, as plotted), a fraction of the cross-encoder's (§5.6). Per event it is a one-way bet on acquirer-renaming, and there it is genuinely hard to beat: on the M&A slice of the TIME split (n=1,506) it scores 0.741, *above* both the L-6 (0.625) and the ensemble (0.711) — the event type the inversion is built for (full-record models still outscore it there). The bet is one-way: the same heuristic scores 0.11 on pure renames and 0.35 on legal-form changes, where the cross-encoder scores 0.90 and 0.85, and it loses pairwise AUROC to the cross-encoder on every split (Table 1). Curated-pool Hits@1 alone can therefore be gamed; within the name-input class, the cross-encoder's claim is being strong *simultaneously* on the curated pool, the alternative-miner pool, the

Table 1: Test-set metrics on the three split regimes, released pool. Bold = headline L-6 cross-encoder used in baseline comparisons; the L-12 matches or beats it throughout, and the L-6×L-12 ensemble (Table 2) is the strongest *name-input* method that survives pool shift (§5.4). Random ranking scores 0.18–0.22 Hits@1 (pools average 7.4–8.3 candidates); the best bi-encoder lifts Hits@1 5–8 pp over zero-shot E5-small and the cross-encoder a further 24–35 pp. Hits@1 95% CIs: ±2.0 / ±2.3 / ±8.5 pp. *Supervised GBMs (§5.4): the name-feature variant and `anti-token_set` (*negated* similarity) exploit the construction — the strongest name-input rows here, near-chance off-pool — while `features_gbm-full`, trained on the cross-encoder's negatives with the full structured record, transfers across all three pools (Figure 2), as does the full-record cross-encoder — the strongest fielded system in the paper (§5.4). Protocol and metrics: §5. †Opus: no temperature param; Gemini (preview): thinking not disableable, lowest level, Batch API. ‡local; 56/52/50-anchor sample.

	Held-out time			Held-out event			Held-out jurisdiction		
Method	AUROC	Hits@1	MRR	AUROC	Hits@1	MRR	AUROC	Hits@1	MRR
`exact_name`	0.474	0.163	0.373	0.523	0.193	0.400	0.571	0.352	0.539
`token_set`	0.193	0.136	0.319	0.266	0.180	0.356	0.323	0.356	0.522
`tfidf_char`	0.277	0.180	0.360	0.349	0.213	0.386	0.568	0.420	0.583
`sentence_encoder`	0.315	0.204	0.385	0.386	0.232	0.408	0.648	0.432	0.587
`anti-token_set` (1−Jaccard)	0.807	0.700	0.761	0.734	0.638	0.715	0.677	0.516	0.618
`features_gbm` (supervised)*	0.935	0.772	0.840	0.926	0.734	0.816	0.960	0.835	0.897
`features_gbm-full` (full record)*	0.929	0.783	0.860	0.912	0.753	0.838	0.893	0.881	0.924
LLM pairwise, zero-shot — uniform single-pair protocol, full test coverage									
`Claude Haiku 4.5`	—	0.290	0.476	—	0.335	0.517	—	0.618	0.738
`Claude Sonnet 4.6`	—	0.301	0.492	—	0.364	0.542	—	0.672	0.758
`Claude Opus 4.7`†	—	0.350	0.534	—	0.416	0.589	—	0.767	0.839
`Gemini 3.1 Pro`†	—	0.235	0.433	—	0.278	0.474	—	0.569	0.699
`DeepSeek V4 Pro`	—	0.260	0.451	—	0.309	0.497	—	0.585	0.712
`gpt-oss-20b`‡	—	0.185	0.377	—	0.203	0.403	—	0.523	0.683
`method_full` (bi-enc, token-overlap negs)	0.403	0.255	0.425	0.451	0.291	0.450	0.667	0.500	0.627
`method_no_hard` (bi-enc, no explicit negs)	0.407	0.259	0.429	0.461	0.299	0.459	0.668	0.511	0.641
`method_v2_full` (bi-enc, E5-cosine negs)	0.405	0.250	0.424	0.457	0.296	0.455	0.694	0.500	0.629
Cross-encoder (L-6, 22M)	**0.836**	**0.607**	**0.725**	**0.825**	**0.569**	**0.701**	**0.868**	**0.750**	**0.818**
Cross-encoder (L-12, 33M)	0.844	0.631	0.742	0.836	0.598	0.724	0.870	0.750	0.823
Cross-encoder (L-6, full record)*	0.930	0.761	0.845	0.915	0.708	0.808	0.954	0.875	0.916

Table 2: Cross-encoder extensions: Hits@1 [95% bootstrap CI]; L-6 row repeated from Table 1. Prompt enrichment is the largest increment on TIME/EVENT; the L-12 pairing captures the JURIS gain (CIs overlap — read per-step deltas as directional).

Model	TIME	EVENT	JURIS
Cross-encoder (L-6, 22M)	0.607 [0.587, 0.627]	0.569 [0.546, 0.592]	0.750 [0.665, 0.835]
+ enriched prompt	0.675 [0.653, 0.697]	0.642 [0.620, 0.662]	0.750 [0.659, 0.830]
+ Wikidata-LEI positives	0.694 [0.669, 0.712]	0.651 [0.630, 0.673]	0.773 [0.682, 0.853]
+ L-12 backbone (33M)	0.697 [0.678, 0.716]	0.676 [0.657, 0.697]	0.761 [0.670, 0.841]
Ensemble L-6×L-12	**0.700 [0.681, 0.718]**	**0.677 [0.656, 0.700]**	**0.784 [0.682, 0.864]**

fielded pool, and every event type with a usable sample (Figure 2). A plausible mechanism is joint attention: asking whether the *differences* between two records look like a lifecycle-event pattern, not how similar two strings are.

5.5 Per-event-type breakdown

Aggregate Hits@1 hides large per-event-type variation.

Held-out time. Pure renames (`CHANGE_LEGAL_NAME`, n=10) are essentially solved by every similarity-direction method including zero-shot E5-small (0.90 Hits@1, no lift); only the inverted exploit fails them (0.11, §5.4). The headline gains come from the events where the post-event name comes from the acquirer and string similarity cannot bridge the gap: `MERGERS_AND_ACQUISITIONS` (n=1,506: 0.18 zero-shot → 0.62 L-6 → 0.71 ensemble; the inverted exploit's best slice, 0.74, §5.4) and `ABSORPTION` (n=461: 0.24 → 0.54 → 0.65). The ensemble also more than doubles zero-shot on the small but operationally critical `BREAKUP` subset (n=7: 0.14 → 0.71).

"Held-out event type" really means "held-out absorption." The held-out-event test set is 94.5% `ABSORPTION` anchors (§3.4); `DEMERGER`, `BREAKUP`, and `SPINOFF` contribute n=42, n=20, and n=4 within the Hits@1-eligible subset. Table 3 shows the lift is consistent across the absorption-like types and that the `SPINOFF` cell is statistically uninformative, so the aggregate "+34 pp Hits@1 over zero-shot E5" on this split is primarily a transfer claim about absorption, not a strong claim about spinoffs.

5.6 Robustness to candidate-pool shift

The headline cross-encoder result is trained and (mostly) evaluated against candidates from the same family of hard-negative miners — v1 token-overlap and v2 E5-cosine both rank candidates by name-related signals, and the cross-encoder trains on E5-cosine negs. The obvious question: does the gap survive an evaluation pool built

Table 3: Per-event-type Hits@1 on the held-out-EVENT test split (Hits@1-eligible subset), zero-shot E5-small vs. L-6. The full test split is 94.5% `ABSORPTION` (2,609 of 2,760); `SPINOFF` at n=4 is statistically uninformative. Lift concentrates in the absorption-like types.

Event type	*n* anchors	zero-shot E5	Cross-enc (L-6)	Δ (pp)
ABSORPTION	1,546	0.230	**0.571**	+34.1
DEMERGER	42	0.333	**0.595**	+26.2
BREAKUP	20	0.150	**0.400**	+25.0
SPINOFF	4	0.500	0.250	−25.0*

* n=4, statistically uninformative.

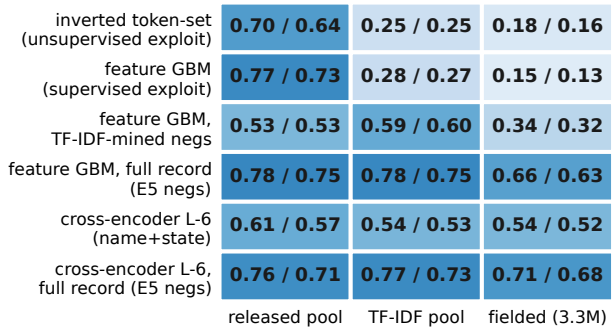


Figure 2: The argument in one exhibit: expected Hits@1 (TIME / EVENT; fill = mean) as the candidate pool shifts from the released pool to an unrelated miner's pool to retrieval from the full 3.3M-entity universe. Read rows by their *minimum*. The name-only exploits (rows 1–2) collapse off-pool; the name-feature GBM retrained on TF-IDF negatives (row 3) recovers only on its own miner's pool. Among name-input methods only the cross-encoder transfers (row 5). Given clean negatives *and* the full record, trees (row 4) and joint encoding (row 6) both transfer — and the full-record cross-encoder holds the highest minimum and wins the fielded pool (§5.4). The name+state TF-IDF cell is the `trn-E5` reproduction (within 2 pp); fielded cells include the gold (strict conditional and end-to-end in the text); JURIS in the text and tables.

with a mechanically different miner, or is the headline number partly an artifact of how the negatives were chosen?

Setup. We re-mine hard negatives with sparse char-3gram TF-IDF cosine — no learned representation, no shared parameters with E5 — holding the other three classes, splits, and positives constant (Sylvester-TF-IDF). We train cross-encoders on the E5-cosine negatives (trn-E5) and the TF-IDF negatives (trn-TF-IDF) and score each on both pools. The released pool's negatives are token-overlap-mined, so the headline configuration is itself already cross-miner; the only miner-matched cell is trn-TF-IDF on the TF-IDF pool.

What the 2×2 tells us. The architecture generalizes across miners: each trained model lands within 1–2 pp of the other on the pool it never saw (TIME/EVENT), miner-matching is worth a bounded +7–11 pp, and the gain over zero-shot E5 survives every pairing (+13 to +40 pp, outside any CI band) — though its size depends on the miner, so read the abstract's "+24–35 pp" in that light.

Fielded retrieve-then-rerank. The miner-robustness experiment above shifts the mining signal but still draws candidates from a curated negative pool. A real deployment pipeline does something different: a first-stage retriever pulls the top-K candidates from the full entity universe, then a second-stage model reranks. We simulate that here. The trained `method_v2_full` bi-encoder retrieves top-20 candidates per test anchor from the full 3.3M-entity GLEIF universe, restricted to the anchor's jurisdiction (the lightest filter a production system would apply); the cross-encoder reranks. The filter is nearly free (1.8–4.1% of test positives cross countries, capping recall at ≥95.9%, far above the observed 33–50%): the bottleneck is the embedding, not the filter.

Bi-encoder Recall@20 is 33.2 / 39.6 / 49.7% on TIME / EVENT / JURIS. That is the hard ceiling for any reranker. When the bi-encoder does retrieve the true successor, the L-6 picks it #1 with 46–57% Hits@1, the ensemble 53–60%, the full-record GBM 51–66% (n.s. vs. the ensemble) — and the full-record cross-encoder 64–78%, +12 pp clear of every other reranker ($p < 10^{-4}$; §5.4). End-to-end deployment Hits@1 (recall × conditional, un-retrieved truths counted as misses) is 15–28% for the L-6, 17–30% for the ensemble, 17–33% for the full-record GBM, and 21–39% for the full-record cross-encoder. The drop from in-distribution to deployment confirms rather than contradicts §4.2: the bi-encoder's fixed-embedding limit, which fails to retrieve lifecycle-spanning successors by cosine similarity, is exactly the bottleneck the anti-correlation analysis predicts.

5.7 Ablations

State features help modestly. Ablating jurisdiction and legal form from the prompt costs 1.1–1.8 pp Hits@1 on every split, most on held-out event type and least on held-out jurisdiction.

Explicit hard negatives slightly hurt the bi-encoder. (Table 1): `method_no_hard` edges out the v1 full model by 0.4–1.1 pp Hits@1 on every split, and E5-cosine re-mining does not change that. The culprit is the `token_overlap_same_country` class, 92% of mined v1 negatives and the lowest-precision one; in-batch negatives from MultipleNegativesRanking are cleaner on their own. So better explicit negatives do not always help contrastive training — but they are exactly what the cross-encoder needs (§4.2).

5.8 Additional diagnostics

Four diagnostics; full tables ship with the released code. **B-cubed cluster F1** [2] is 0.92 / 0.90 / 0.91 for the name+state cross-encoder and 0.94 / 0.93 / 0.94 for the full-record one, vs. ≤0.48 for any bi-encoder (which collapse to one giant cluster at the tuned threshold); restricted to true clusters of size ≥2, a more honest 0.85 / 0.84 / 0.88 (name+state). **Calibration and selective prediction.** ECE (10-bin) 0.20, Brier 0.18 for the name+state cross-encoder (0.13 / 0.12 full-record) — over-confident in the high-score bin; a train-fitted temperature does not fix it, so we report uncalibrated scores. A selective-prediction [5] sweep on the ensemble reaches

Table 4: Hits@1 with 95% bootstrap CIs for the 2×2 cross-encoder train/evaluation matrix; row labels name the training miner. The zero-shot E5 row is bootstrapped separately (within 0.002 of Table 1). The advantage over E5 holds in headline pairings (+20 to +40 pp) and swapped (+13 to +39 pp); re-trained `trn-E5` cells reproduce the original within 2 pp on time/event.

	Eval on benchmark pool (`pairs.parquet`)			Eval on TF-IDF pool (`pairs_v3.parquet`)		
	TIME	EVENT	JURIS	TIME	EVENT	JURIS
E5-small zero-shot	0.204 ±1.8	0.232 ±2.1	0.432 ±10.5	0.265 ±1.4	0.320 ±1.7	0.441 ±7.8
cross-enc trn-E5	**0.601** ±2.2	**0.552** ±2.4	0.705 ±8.8	0.544 ±1.8	0.529 ±1.7	0.571 ±7.5
cross-enc trn-TF-IDF	0.589 ±2.0	0.534 ±2.4	**0.750** ±9.1	**0.655** ±1.5	**0.615** ±1.7	**0.640** ±7.1
Δ *vs. E5, headline pairing*	+39.7	+32.0	+27.3	+39.0	+29.5	+19.9
Δ *vs. E5, swapped pairing*	+38.5	+30.1	+31.8	+27.9	+20.9	+13.0

Table 5: Negative-class mean score (lower is better) on negatives *never seen in training* (n = 99 / 1,398 / 1,096 / 20,759): *p/s* parent/subsidiary, *sn-D* same-name/diff-country, *sn-S* same-name/same-country, *tok* token-overlap. The ensemble improves three classes but *worsens p/s* (0.67 → 0.75): prompt-injected L2 context makes parent/subsidiary pairs look *more* similar. Scores are on each method's own scale (compare down columns); bold = lowest among learned methods.

Method	p/s	sn-D	sn-S	tok
`tfidf_char`	0.220	0.590	0.783	0.595
`sentence_encoder`	0.841	0.912	0.960	0.905
`method_full`	0.501	0.438	0.893	0.747
Cross-encoder (L-6, 22M)	0.668	0.201	0.432	0.311
Ensemble (L-6×L-12)	0.748	**0.166**	**0.358**	**0.284**

Hits@1 0.93–0.98 at 20% coverage and 0.83–0.98 at 50%, so high-precision users can threshold. **Cross-domain transfer.** Zero-shot Magellan Walmart-Amazon [6]: the cross-encoder *loses* to its own zero-shot base encoder by 5.9 pp AUROC. The fine-tuned representation is specialized to lifecycle-spanning legal-entity matching, not generic ER. **Label quality, second opinion.** A rule-based re-classifier — orthogonal to the primary's event-type evidence, though still name-informed — agrees with it on 98.7% [95% CI 97.0–100%] of the 231/277 stratified-sampled positives it judges confidently (46 marked uncertain), bounding heuristically detectable label noise below 2% there; human expert review is future work.

5.9 Where each method is fooled

Table 5 reports per-negative-class mean score (higher = more often fooled). Bi-encoders cut zero-shot E5's score on `parent_subsidiary` and `same_name_diff_country` by 40–52%, but `same_name_same_country` remains a wall at ≈0.89. The cross-encoder breaks it (0.89 → 0.43; 0.36 ensemble) and nearly solves the other name classes (0.91 → 0.17 / 0.28). The one regression is `parent_subsidiary` (0.50 → 0.67 → 0.75 across bi-encoder, L-6, ensemble): the prompt enrichment that injects GLEIF L2 parent/child names makes it *worse*. At 99 held-out examples the headline impact is small, but the direction is clear: explicit multi-class modeling of corporate-hierarchy relationships.

6 ANALYSIS

6.1 Error analysis

Of 100 sampled `method_full` failures on held-out time: 66 mixed/other, 19 where anchor and true successor share *no* tokens, 8 substring-overlap distractors, 7 exact normalized-name collisions. The no-shared-token cases are the hardest for any name-input method — they need world knowledge, graph context, or the full record (§5.4); Figure 1's anchor is representative: resolving "DB" to Deutsche Bank from name strings alone is beyond what a 33M encoder reliably does.

Residual JURIS failures cluster on sibling consolidation. Of the 19 / 88 anchors the ensemble mis-ranks on JURIS, ≥11 are sibling consolidations (7 are "`EPIC <port> PTE. LTD.`" vessels absorbed into one owner; the model returns sibling vessels), plus Japanese-script renames the ASCII tokenizer cannot bridge. Multilingual pretraining and ownership-graph context would address both.

6.2 Why this matters in practice

Merchant embeddings at scale [11], vendor deduplication, and compliance entity backbones all face the lifecycle-ER gap after a merger or rebrand; Sylvester lets practitioners test for it — and §5.4 says the cheapest fix is feeding the matcher the whole record.

7 LIMITATIONS AND FUTURE WORK

The "anti-correlation" finding is a construction artifact. of mining name-similar negatives against name-different positives (§5.4); it motivates joint encoding here, not a universal claim about name similarity in ER.

In-distribution headline overstates deployment. The headline Hits@1 is measured on the released pool; end-to-end on the fielded pipeline every reranker falls to 15–39% (§5.6) because bi-encoder Recall@20 is 33–50%, so the reranker often never sees the true successor. The gain is partly pool-specific but survives miner shift (+13 to +39 pp; Table 4). A stronger first stage (hybrid sparse–dense, or a pre-filtered shortlist) is the highest-leverage future work.

Seed variance and a sliver of mislabeled hard negatives. Over seeds 42/7/1337 the bi-encoder varies ±0.5/0.4/1.6 pp and the enriched cross-encoder ±0.5/0.3/1.6 pp (TIME/EVENT/JURIS). The new headline systems are stable too: the full-record GBM moves ≤0.5 pp on-pool and ≤1.2 pp fielded-conditional; the full-record cross-encoder spreads 1.6–3.4 pp and 3.3–3.9 pp respectively, and its *worst* seed still leads the GBM's best fielded by ≈8 pp. Separately, 205/93,695 = 0.22% of hard negatives appear in GLEIF's `SuccessorEntity`; dropping them moves results ≤0.2 pp.

Survivorship bias and thin slices. GLEIF `SuccessorEntity` is voluntarily populated, biasing positives toward well-counseled regulated entities; of 302 marquee Wikidata M&A pairs, 283 are absent from it (§3.4), so high-profile events are under-represented. The held-out-event test set is 94.5% absorption and the jurisdiction test has 88 eligible anchors (±8.5 pp CI): gains clear the CI band on all three splits, but per-split absolutes carry those sample sizes. Eligibility itself conditions on a name-colliding neighbour existing, amplifying the construction (§5.2); a 277-pair second-classifier check bounds detectable label noise at <2% (§5.8). Scoring the 244 released marquee pairs against *random* same-country pools (no adversarial mining; script released) inverts the picture: token Jaccard scores 0.41 while the name+state cross-encoder falls below chance (0.07) — models trained here inherit the pool's anti-similarity prior — and the full-record models recover only partly (0.21–0.24). Rankers trained on mined pools should not be assumed to transfer to unmined candidate distributions. Splits are pair-level, not entity-disjoint (32% / 21% of time-split test candidates / gold successors appear in training pairs; JURIS <1%), though L-6 Hits@1 shows no seen-entity advantage (unseen 0.62 vs. seen 0.57 TIME; 0.56 vs. 0.58 EVENT, both n.s.); entity-disjoint splits are a stricter future design.

LLM baselines: what the discrete scale hides. A 0–10 rubric caps what ranking can express: models forfeit 13–48 pp Hits@1 to within-pool ties, and Gemini 3.1 Pro scores every candidate identically on 55% of TIME anchors — oracle tie-breaking would give it 0.72 vs. 0.23 measured, above our L-6 — while Sonnet/Opus collapse only ≈5% of pools and the cross-encoder's continuous scores pay no tie tax. Prompt format also moves results (batched Sonnet 4.6 scores 0.46 TIME vs. 0.40 single-pair on the same 56 anchors). The panel therefore bounds what this elicitation format extracts, not what these models can do: finer scales, log-probs, or comparative tie-breaks would likely raise them — as would a full-record prompt, the LLM analogue of §5.4's finding — and cross-vendor gaps may partly reflect exposure to GLEIF-derived text.

The model is specialized, and parent/subsidiary is a thin slice. The cross-encoder underperforms zero-shot E5-small on out-of-domain product matching (Walmart-Amazon, −5.9 pp AUROC; §5.8): the representation is specialized to lifecycle ER, not generic. The full-record cross-encoder closes the input gap (§5.4) but pays the usual joint-encoding inference premium over trees; a full-record *LLM* prompt is the remaining untested cell. Parent/subsidiary negatives regress (0.50 → 0.67 → 0.75 across bi-encoder, L-6, and ensemble), though at 99 held-out examples the effect is weak; explicit multi-class modeling of related-but-distinct entities is clear future work.

8 CONCLUSION

Sylvester is an open benchmark for lifecycle-aware entity resolution on financial entities, built entirely from public GLEIF, SEC EDGAR, and Wikidata data, with a three-pool protocol that separates methods that learn the pool from methods that learn the task. Name-only exploits post the paper's best name-input scores and collapse off-pool. Among name-input methods, one 22M-parameter cross-encoder is the only survivor, above every zero-shot LLM we test on the two large splits. The input sets the ceiling: boosted trees over the full structured record transfer across every pool at negligible cost, and a cross-encoder over the same record keeps the most signal under shift, winning the fielded universe by 12 pp — the lifecycle signal lives in the record, and joint encoding is how you keep it. What remains open is the first stage: even the best reranker is capped by 33–50% retrieval recall, so in our fielded simulation it is candidate generation, not matching, where lifecycle ER breaks.

Reproducibility. Code (MIT), data (CC-BY-4.0), parsers, predictions, trained weights, and a datasheet: https://github.com/chang-07/er-research; archival DOI on acceptance. Every non-LLM result reproduces from raw GLEIF in ≈2 hours on one MacBook.

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